Second Partial

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Critical Points

What are the critical points?

Critical points are the roots of the derivative of a funciton or the values of the domain in which the derivative is undefined.

Why we are interested in critical points?

Critical points are essential for identifying potential local maxima, minima, or inflection points in the graph of the function.

Example: $f(x) = e^{\cos(x)}$

The derivative of the function $f(x) = e^{\cos(x)}$, by using the chain rule, is

$$f'(x) = -e^{\cos(x)} \sin(x).$$

Then to compute find the critical points we need to equate the derivative to zero and solve for x,

$$f'(x) = 0 \leftrightarrow -e^{\cos(x)} \sin(x) = 0$$

Example: $f(x) = e^{\cos(x)}$

To solve $-e^{\cos(x)}\sin(x) = 0$ we have two possible solutions:

$$e^{\cos(x)} = 0, \quad \sin(x) = 0$$

$\sin(x) = 0$

The solutions of this equation are,

$$x = n\pi,\ n \in \mathbb{Z}$$

$e^{\cos(x)} = 0$

$$\cos(x) = \ln(0)$$

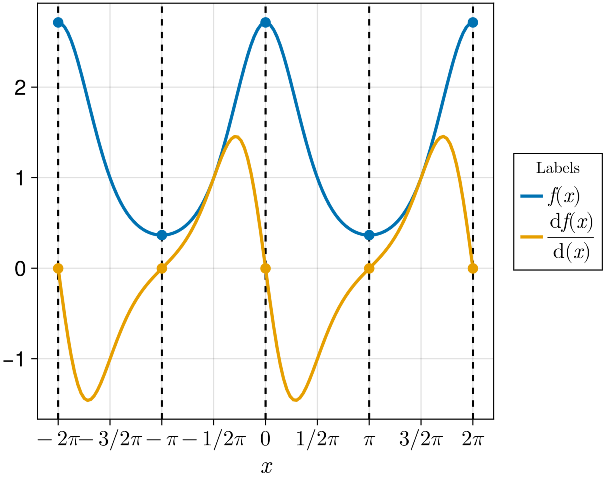
No solution.

Example: $f(x) = e^{\cos(x)}$

Since we have only solution for $\sin(x) = 0$ the critical points of the function $f(x)$ are,

$$\boxed{x = n\pi,\ n \in \mathbb{Z}}$$

Example: $f(x) = e^{\cos(x)}$



Class activity

Compute the derivative of the following function. Then find its critical point.

$$P(x) = x100e^{-0.05x} - 20 \cdot 100e^{-0.05x} - 500$$